SENSOR REPORT

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***Abstract*— This report discusses the working principles and applications of sensors used in autonomous vehicles, including LiDAR, cameras, IMU, GNSS and wheel speed sensors. It also covers sensor fusion, onboard processing and electronic steering and braking systems, highlighting their role in vehicle perception, navigation and control.**

***Keywords*— Autonomous Vehicles, Sensors, LiDAR, IMU, GNSS, Sensor Fusion, Vehicle Control.**

INTRODUCTION

A driverless Formula Student car relies on a set of sensors, computing hardware and control systems to understand the track and move safely without a human driver. The sensors provide information about the surroundings and the vehicle's own motion. Software processes this information to detect track features, estimate the vehicle's position, plan a path and send commands to the steering and braking systems. The performance of the complete system depends not on one sensor alone, but on how reliably the different components work together.

The sensor system supports the main stages of autonomous driving: perception, localization, tracking, prediction, planning and control. Perception identifies relevant features such as cones and other objects. Localization estimates where the car is and which direction it is facing. Tracking follows detected objects over time, while prediction estimates how their positions may change. The planning system uses this information to choose a suitable path and speed. Finally, the control system converts the planned motion into steering, acceleration and braking commands.

HESAI PANDAR40P LiDAR

The Hesai Pandar40P is a 3D LiDAR sensor manufactured by Hesai Technology. It is used in driverless vehicles to detect and measure the distance of objects in the surrounding environment. The sensor has 40 laser channels and rotates mechanically to provide a 360° horizontal field of view and a 40° vertical field of view, covering angles from −25° to +15°. Its range can reach up to 200 metres under specified conditions, and it operates at 10 or 20 Hz, depending on the configuration. The multiple laser channels scan at different vertical angles, allowing the sensor to capture different parts of an object, such as its body, windows and roof.

The Pandar40P works using the Time of Flight (ToF) principle. It emits laser pulses and measures the time taken for the reflected light to return from an object. By measuring distances in many directions, the sensor generates a 3D point cloud, which is a collection of points representing the surrounding environment. Each point provides spatial information that helps the vehicle understand the position and shape of objects such as cones, other vehicles and track boundaries. Unlike a conventional camera, LiDAR does not inherently provide RGB colour information, but it directly measures distance and provides useful information about the geometry of the environment.

The raw point cloud must be processed before it can be used for autonomous driving. First, unnecessary points outside the required region are cropped, and the cloud may be downsampled using a voxel grid filter to reduce the number of points and processing time. Noise and isolated points are removed using statistical or radius-based outlier removal. Points belonging to the vehicle itself are also removed so that the system does not detect parts of its own body as obstacles. The next step is ground removal, which separates road or track points from objects above the ground. The remaining points are grouped into clusters using methods such as Euclidean clustering with a KD-tree or DBSCAN. These clusters can then be used to estimate object boundaries, classify objects and track them across consecutive scans. Tracking helps the system maintain an object's identity and estimate its movement, even when it is temporarily hidden

Sourcing and cost :Hesai Pandar40P, a 40-channel mechanical LiDAR providing 360° scanning. Its cost generally exceeds ₹3,00,000, and it is usually procured through authorized distributors or industrial import channels.

VECTORNAV 20

The VectorNav VN-200 is a compact GNSS-aided Inertial Navigation System (GNSS/INS) used to estimate a vehicle's position, velocity and orientation. It combines satellite navigation with measurements from inertial sensors and a magnetometer to provide information about the vehicle's movement and direction. GNSS stands for Global Navigation Satellite System, which includes systems such as GPS and Galileo. The GNSS receiver uses satellite signals to estimate the vehicle's position and velocity, while the inertial sensors provide frequent measurements of motion between satellite updates.

The VN-200 contains three main types of sensing components: accelerometers, gyroscopes and a magnetometer, along with a GNSS receiver. The accelerometers measure acceleration along three axes, helping the system understand how the vehicle speeds up, slows down or changes direction. The gyroscopes measure angular velocity, showing how quickly the vehicle rotates around its axes. The magnetometer measures the surrounding magnetic field and can help estimate the vehicle's heading. Together, these measurements provide information about the vehicle's motion and orientation.

The VN-200 uses an onboard navigation filter, based on sensor-fusion algorithms, to combine the different measurements and estimate the vehicle's position, velocity and attitude. GNSS helps correct the errors that gradually accumulate in inertial estimates, while the inertial sensors provide frequent motion updates between GNSS measurements. However, GNSS accuracy can be affected by signal obstruction, and magnetometer readings can be disturbed by nearby metal structures, motors and electrical currents.

The VectorNav VN-200 uses an Extended Kalman Filter (EKF) to combine data from its Inertial Measurement Unit (IMU) and Global Navigation Satellite System (GNSS) receiver to obtain accurate navigation estimates. The accelerometers and gyroscopes continuously measure the vehicle's acceleration and angular motion, while GNSS provides position and velocity measurements. The Kalman filter predicts the vehicle's motion using IMU data and corrects these predictions using GNSS measurements, accounting for sensor noise and measurement uncertainty. This sensor-fusion process helps estimate the vehicle's position, velocity, and orientation while reducing the effects of accumulated inertial errors. In a driverless car, the Kalman filter is important for reliable localization and navigation, particularly when individual sensor measurements are noisy or temporarily unreliable

Sourcing and cost: When imported through Indian engineering distributors, the price typically ranges between ₹3,000,000 to ₹4,50,000 INR per unit for standard requirements. This variance includes international freight, customs duties, and local dealer margins.

STEREO ZED 2I CAMERA

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The Stereolabs ZED 2i is a stereo camera used in driverless vehicles to detect objects, estimate their distances, and understand the surrounding environment. It provides both visual information and depth information, making it useful for detecting and identifying track cones in autonomous racing.

The ZED 2i has two cameras positioned a fixed distance apart. Both cameras capture images of the same scene from slightly different viewpoints. Its software identifies matching visual features in the left and right images and calculates the difference in their positions, known as disparity. Using this disparity, the distance between the cameras, and the camera's calibration parameters, the system estimates the object's depth through a geometric method called triangulation. Generally, a larger disparity indicates a closer object, while a smaller disparity indicates a farther object.

The camera provides RGB images, depth maps, and 3D point clouds. RGB images contain colour and appearance information, which helps perception software distinguish cone colours and recognise objects. Depth maps estimate the distance of pixels in an image, while 3D point clouds represent objects and their surroundings in three-dimensional space. The ZED 2i also contains an IMU, including an accelerometer and gyroscope, as well as a magnetometer. Its visual-inertial tracking software combines visual features with inertial measurements to estimate the camera's motion and orientation, supporting visual-inertial SLAM

ZED 2i can help identify cones, distinguish their colours, estimate their positions and contribute to track-boundary detection. It complements the Hesai Pandar40P LiDAR: the camera provides colour and visual-recognition information, while LiDAR provides range measurements and detailed geometric information. Combining the two sensors can improve environmental perception, although accurate calibration, time synchronization and coordinate transformations are necessary to align their measurements correctly.

One limitation of stereo vision is that depth estimation depends on finding matching features in both images. Poor lighting, glare, reflective or textureless surfaces, and occlusion can reduce reliability. Depth accuracy also generally decreases as distance increases. Therefore, combining the camera with LiDAR can provide more reliable environmental information than relying on either sensor alone.

Sourcing and cosr : The Stereolabs ZED 2i is a stereo camera used for depth estimation, object detection and visual perception in autonomous vehicles. It can be purchased through Indian suppliers such as MG Super Labs and Fab.to.Lab. The estimated cost is approximately ₹73,000, depending on availability and supplier pricing. .

NVIDIA Jetson AGX Orin

The NVIDIA Jetson AGX Orin is a high-performance embedded computing platform designed to run artificial intelligence and other computationally demanding applications. In a driverless car, it acts as the onboard computer responsible for processing data received from different sensors and running the software required for autonomous driving. Unlike sensors such as LiDAR and cameras, which collect information about the surroundings, the Jetson AGX Orin processes this information and helps the vehicle understand its environment and make driving decisions.

The Jetson AGX Orin consists of a Central Processing Unit (CPU), Graphics Processing Unit (GPU), memory and interfaces for connecting external devices. The CPU handles general instructions, program execution and coordination between different software components, while the GPU performs multiple calculations simultaneously. This parallel processing capability is particularly useful for running artificial intelligence models and processing large amounts of image and sensor data. The platform also includes dedicated AI acceleration capabilities that support computationally intensive tasks such as object detection and environmental perception.

In an autonomous vehicle, the Jetson AGX Orin receives

data from sensors such as LiDAR, cameras and navigation systems through suitable communication interfaces and software drivers. It runs perception algorithms to process this data and identify important features in the surroundings, such as traffic cones, obstacles and track boundaries. Information obtained from different sensors can also be combined to develop a more accurate understanding of the vehicle's environment. The processed information is then passed to the path-planning software, which determines a suitable route for the vehicle. Based on this planned path, the control system generates the necessary steering, acceleration and braking commands.

One of the main advantages of the Jetson AGX Orin is its ability to handle multiple sensor data streams and execute several applications simultaneously. Its GPU and AI accelerators make it suitable for tasks that require fast processing, particularly computer vision and neural network inference. However, its performance depends on how efficiently the software is developed and how well the available computing resources are managed. Running several demanding applications can increase power consumption and heat generation, making proper cooling and power management necessary. The system must also process sensor information with minimal delay, since delayed results can affect the vehicle's ability to respond to changes in its surroundings.

Sourcing and cost : The NVIDIA Jetson AGX Orin is an embedded computing platform used to process sensor data and execute artificial intelligence algorithms in autonomous vehicles. It supports tasks such as object detection, sensor fusion and path planning. According to the reference report, it can be sourced through distributors serving India, including Arrow Electronics and Mouser India, at an estimated cost of ₹4,00,000–₹5,50,000 per unit.

INERTIAL NAVIGATION SYSTEM

An Inertial Navigation System (INS) is a navigation system used to estimate a vehicle's position, velocity and orientation by continuously monitoring its motion. It uses measurements obtained from an Inertial Measurement Unit (IMU), which consists of accelerometers and gyroscopes. The accelerometers measure the acceleration of the vehicle, while the gyroscopes measure its rotational motion. By processing these measurements over time, the INS estimates how the vehicle moves, how fast it is travelling and how its direction changes.

In a driverless car, the INS plays an important role in localization and motion estimation. It helps the vehicle determine its position and orientation while moving along a track. The system calculates changes in velocity and position using the acceleration and rotational measurements provided by the IMU. However, since the INS calculates position rather than measuring it directly, even small errors in the sensor readings can accumulate over time. This gradual accumulation of errors is known as inertial drift, which can cause the estimated position or orientation to differ from the vehicle's actual state.

To reduce these errors and improve navigation accuracy, an INS is often combined with a Global Navigation Satellite System (GNSS). GNSS provides satellite-based position measurements that help correct the errors accumulated by the inertial sensors. At the same time, the INS provides frequent motion estimates between GNSS updates. This combination allows the vehicle to maintain more reliable estimates of its position, velocity and orientation.

INERTIAL MEASUREMENT UNIT

An Inertial Measurement Unit (IMU) is an electronic sensor device used to measure the acceleration and rotational motion of a vehicle. It mainly consists of two types of sensors: accelerometers and gyroscopes. The accelerometers measure acceleration along three axes, while the gyroscopes measure the rate of rotation around these axes. Some IMUs also include a magnetometer, which measures the surrounding magnetic field and can assist in determining the vehicle's heading
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In a driverless car, the IMU continuously provides information about the vehicle's movement and changes in orientation. The accelerometer detects changes in acceleration as the vehicle speeds up, slows down or changes direction, while the gyroscope measures how quickly the vehicle rotates. These measurements are useful for understanding the vehicle's motion and are supplied to navigation systems for further processing.

The IMU itself does not directly determine the vehicle's position. Instead, it provides the motion measurements required by systems such as the INS to calculate position, velocity and orientation. Since small errors in accelerometer and gyroscope measurements can accumulate when they are used to estimate motion over time, IMU data is often combined with information from other sensors, such as GNSS, to improve navigation accuracy.

The main difference between an IMU and an INS is that an IMU measures acceleration and rotational motion, whereas an INS processes these measurements to estimate the vehicle's navigation state. Therefore, the IMU acts as a source of motion data, while the INS uses that data to estimate how the vehicle is moving and where it is located

Sourcing and cost : The Bosch BMX160 9-axis IMU, which integrates accelerometer, gyroscope and magnetometer measurements, costs approximately ₹3,000 and is available through Robu. The WIT HWT905-485 9DOF AHRS IMU is a mid-range alternative priced between ₹10,000 and ₹50,000. For higher-accuracy navigation applications, the VectorNav VN-200 is an industrial-grade option costing above ₹1,00,000..

WHEEL SPEED SENSOR

A wheel speed sensor is a device used to measure the rotational speed of a vehicle's wheels. It provides information about how fast each wheel is rotating, which can be used to estimate the vehicle's speed, calculate the distance travelled and detect wheel slip. In a driverless car, wheel speed sensors play an important role in vehicle motion estimation, odometry and control. They help the system monitor the movement of the vehicle and provide feedback that can be combined with information from other sensors.

Wheel speed sensors generally work by detecting the movement of teeth or magnetic features on a rotating ring attached to the wheel. A stationary sensor placed near the ring generates electrical pulses as the teeth or magnets pass through its sensing region. The frequency of these pulses

indicates how quickly the wheel is rotating. Common types of wheel speed sensors include Hall-effect sensors, magnetic or variable reluctance sensors, optical encoders and quadrature encoders

The electrical pulses generated by the sensor are collected and processed to calculate the wheel's angular velocity. This can be done using a microcontroller, which counts the pulses over a specific time interval, or by receiving wheel speed information through the vehicle's Controller Area Network (CAN) bus. The angular velocity is calculated using the number of pulses detected, the number of pulses produced per wheel revolution and the time interval. Once the angular velocity is known, the linear speed of the vehicle can be estimated by multiplying it by the effective radius of the wheel, provided there is no significant wheel slip. The sensor readings are also recorded with timestamps so that they can be synchronized with measurements from other sensors.

Before the measurements are used, the data may be filtered to reduce noise and sudden fluctuations. Low-pass filters or moving averages can be applied to obtain smoother speed estimates. The processed wheel speed information can then be used for wheel odometry, which estimates the distance travelled by the vehicle based on wheel rotation. This information is particularly useful for tracking the vehicle's movement between updates from other navigation sensors.

One of the main limitations of wheel speed sensors is that wheel rotation does not always represent the vehicle's actual movement. During wheelspin or skidding, the wheels may rotate faster or slower than expected relative to the vehicle's motion, resulting in inaccurate speed and distance estimates. Measurements can also become less consistent at very low speeds because fewer pulses are generated within a given time interval. To overcome these limitations, wheel speed measurements are often combined with data from an Inertial Measurement Unit (IMU), GNSS or other navigation sensors. This process helps improve motion estimation and allows the vehicle to identify possible wheel slip.

Sourcing and Cost : The ZF (Cherry) GS100502 is an active Hall-effect sensor suitable for measuring wheel rotational speed. It is designed for automotive environments and can be sourced through distributors such as Element14 India or Mouser India. Its estimated cost is approximately ₹3,300 per sensor. A vehicle using one sensor for each wheel would require four units, giving an estimated total of ₹13,200.

SBW

Steer-by-Wire (SBW) is an electronic steering system that uses electrical signals instead of a direct mechanical connection between the steering wheel and the road wheels. In autonomous vehicles, it allows the control software to operate the steering system without human intervention.

The system consists of an Electronic Control Unit (ECU), steering actuator, steering mechanism and feedback sensors. The path-planning software determines the required direction, and the controller calculates the desired steering angle. This command is sent to the ECU, which controls the steering actuator. The actuator converts the electrical signal into mechanical movement, turning the wheels in the required direction.

Steer-by-wire operates using a closed-loop feedback mechanism, where sensors continuously measure the actual steering angle and compare it with the desired angle. Any difference, known as the steering angle error, is corrected by adjusting the actuator's movement. This helps the vehicle follow its planned path accurately.

The main advantage of SBW is precise electronic steering control, making it suitable for autonomous driving. However, it depends on reliable electrical power, communication and electronic components, making redundancy, fault detection and safety mechanisms essential. Steering accuracy can also be affected by tyre grip, vehicle speed and road conditions, even when the commanded steering angle is achieved.

BBW

Brake-by-Wire (BBW) is an electronically controlled braking system that uses electrical signals to control braking instead of relying entirely on mechanical or hydraulic connections. In autonomous vehicles, it allows the control software to regulate speed, slow down and stop the vehicle without human intervention.

The system consists of an Electronic Control Unit (ECU), brake actuator, braking mechanism and feedback sensors. Sensors such as LiDAR and cameras detect obstacles, while the planning and control software determines the required deceleration. The braking command is sent to the ECU, which controls the actuator to generate the necessary braking force. In disc brakes, this force presses the brake pads against the rotating disc, creating friction that slows the vehicle.

Brake-by-wire uses a closed-loop feedback mechanism, where wheel speed, brake pressure and actuator position measurements help the controller monitor and adjust the braking response. It can also work with Anti-lock Braking Systems (ABS) to prevent wheel lock during sudden braking.

The main advantage of BBW is precise electronic braking control, making it suitable for autonomous driving. However, it depends on reliable electrical power, communication and actuators, making fault detection, backup systems and fail-safe mechanisms essential. Braking performance is also affected by tyre grip and road conditions, which can influence stopping distance.

TYPES OF ACCELEROMETERS

An accelerometer is a sensor used to measure the acceleration of an object along one or more axes. Acceleration refers to the rate at which velocity changes with time. An accelerometer typically measures specific force, which includes the effects of motion and gravity. A three-axis accelerometer measures acceleration along the X, Y and Z axes. These measurements can be used to study the vehicle's longitudinal, lateral and vertical motion. Accelerometers are commonly included in an Inertial Measurement Unit (IMU), where they work alongside gyroscopes to provide motion information.

1. Capacitive Accelerometer

A capacitive accelerometer measures acceleration by detecting changes in capacitance between conductive plates.

It generally consists of a movable proof mass placed between fixed electrodes. When acceleration occurs, the proof mass shifts slightly, changing the distance between the electrodes and consequently changing their capacitance. The electronic circuit measures this change and converts it into an acceleration value.

2. Piezoelectric Accelerometer

A piezoelectric accelerometer works on the piezoelectric effect, in which certain materials generate an electrical charge when subjected to mechanical stress. It consists of a seismic mass attached to a piezoelectric material. When acceleration occurs, the mass applies force to the material, producing an electrical charge proportional to the applied force. Piezoelectric accelerometers are particularly suitable for measuring vibrations, shocks and rapidly changing acceleration.

3. Piezoresistive Accelerometer

A piezoresistive accelerometer measures acceleration through changes in electrical resistance caused by mechanical strain. It consists of a proof mass connected to flexible beams containing piezoresistive elements. When acceleration causes the mass to move, the supporting beams deform slightly, changing the resistance of the sensing elements.

4. Servo or Force-Balance Accelerometer

A servo accelerometer uses a closed-loop feedback mechanism to measure acceleration. When acceleration acts on the proof mass, the mass tends to move away from its reference position. A feedback control system detects this movement and applies an opposing electromagnetic or electrostatic force to keep the mass close to its original position.

The force required to maintain this balance is proportional to the acceleration acting on the sensor. Force-balance accelerometers can provide high accuracy and stability, making them useful in precision navigation systems, aerospace applications and scientific instruments. However, they are generally more complex and expensive than conventional MEMS accelerometers.

5. Thermal Accelerometer

A thermal accelerometer measures acceleration by observing the movement of heated gas inside a small enclosed chamber. It contains a heating element and temperature sensors arranged around it. When the device accelerates, the temperature distribution within the chamber changes due to the movement of the heated gas relative to the sensor.

The temperature sensors detect this difference, which is used to estimate acceleration. Since thermal accelerometers do not require a conventional moving proof mass, they can offer good resistance to mechanical shock. However, their response time and measurement performance may be affected by temperature conditions and thermal characteristics.

TYPES OF MAGNETOMETERS

A magnetometer is a sensor used to measure the strength and, depending on its design, the direction of a magnetic field. It is commonly used in navigation systems, electronic compasses, robotics, smartphones and autonomous vehicles. In vehicle navigation, magnetometers are useful for measuring the Earth's magnetic field and assisting in the estimation of the vehicle's heading.

The Earth produces a magnetic field that can be used as a reference for determining direction. A three-axis magnetometer measures the magnetic field components along the X, Y and Z axes. These measurements can be processed to estimate the direction in which the sensor is oriented relative to magnetic north.

Magnetometers are often integrated into IMUs alongside accelerometers and gyroscopes. While accelerometers measure specific force and gyroscopes measure rotational motion, magnetometers provide an external magnetic reference that can help correct heading estimates.

1. Hall-Effect Magnetometer

A Hall-effect magnetometer operates on the Hall effect, which occurs when a current-carrying conductor or semiconductor is placed in a magnetic field. The magnetic field causes moving charge carriers to experience a force, resulting in a voltage difference across the material. This voltage is known as the Hall voltage.

The magnitude of the Hall voltage depends on the magnetic field component perpendicular to the current flow. By measuring this voltage, the sensor can estimate magnetic field strength.

Hall-effect sensors are compact, inexpensive and widely used in automotive applications. They are commonly found in wheel speed sensors, motor position sensors and proximity detection systems. However, conventional Hall-effect sensors generally have lower sensitivity to weak magnetic fields than some other magnetometer technologies**.**

2. Magnetoresistive Magnetometer

A magnetoresistive magnetometer measures magnetic fields by detecting changes in the electrical resistance of a material when it is exposed to a magnetic field. The resistance of the sensing element varies with the strength and direction of the applied magnetic field.

3. Fluxgate Magnetometer

A fluxgate magnetometer is a highly sensitive instrument used to measure weak magnetic fields. It typically consists of a magnetic core surrounded by excitation and sensing coils. An alternating current is applied to the excitation coil, repeatedly driving the magnetic core into saturation.

When an external magnetic field is present, it changes the symmetry of the magnetic response of the core. The sensing electronics detect this change and use it to determine the external magnetic field. Fluxgate magnetometers are known for their high sensitivity and ability to measure weak magnetic fields accurately

4. Induction or Search-Coil Magnetometer

An induction magnetometer operates according to Faraday's law of electromagnetic induction. When the magnetic flux passing through a coil changes, an electromotive force is induced across the coil. The induced voltage is related to the rate of change of magnetic flux. Therefore, search-coil magnetometers are mainly used to detect changing magnetic fields rather than constant magnetic fields.

HALL EFFECT

The Hall effect is the generation of a small voltage across a current-carrying conductor or semiconductor when a magnetic field is applied perpendicular to the current. This voltage varies with the magnetic field strength and direction. Hall-effect sensors are mainly classified into **linear sensors**, which produce a continuous output, and **digital sensors**, which switch states when a magnetic field reaches a certain threshold.

In autonomous vehicles, Hall-effect sensors are used to measure wheel speed, detect steering and pedal positions, monitor motor rotation and measure electric current. They operate without physical contact, work at low speeds and are resistant to dust and vibrations. However, their accuracy can be affected by temperature changes, external magnetic fields and limited sensitivity to weak magnetic fields.

Sourcing and cost : Basic Hall-effect sensors, such as the A3144, are available through Robu for approximately ₹150–₹800. Automotive-grade ABS wheel speed sensors from manufacturers such as Bosch generally cost between ₹3,000 and ₹6,000. These sensors are used for wheel rotation measurement, vehicle speed estimation and wheel-slip detection.

OPTICAL SENSORS

Optical sensors use light to detect objects, movement or changes in position. They generally consist of a light source, such as an LED or laser, and a photodetector that converts received light into electrical signals. They work by detecting changes in the intensity, reflection or interruption of light.

In autonomous vehicles, optical sensors are used for wheel speed measurement, position detection and rotational sensing. Optical encoders detect the movement of a rotating disc containing slots or markings and generate electrical pulses to calculate rotational speed and position. Cameras are also optical sensors and help identify lanes, obstacles, traffic signs and pedestrians.

Optical sensors provide high accuracy, fast response and contactless measurement. However, their performance can be affected by dust, dirt, fog, poor lighting and obstruction of the optical path. Therefore, they require proper positioning and protection to ensure reliable operation.

ERRORS

LiDAR : Vehicle motion during scanning can distort point clouds, resulting in inaccurate object positions. This can be corrected using motion compensation based on VN-200 measurements and wheel odometry. Environmental factors such as rain, dust and unwanted reflections introduce noise, which can be reduced using outlier removal and built-in filtering techniques. Ground detection may become inaccurate on uneven surfaces, requiring zone-based or ring-based ground segmentation with IMU-based tilt correction. Distant cones may generate insufficient points for reliable detection, so a minimum point threshold or camera verification is necessary. Additionally, blind spots and reflections from the vehicle's body can be handled through appropriate sensor positioning and point-cloud masking. Incorrect mounting height or orientation can be identified by comparing the estimated ground plane with the sensor's actual mounting position.

Camera : Stereo cameras become less accurate at estimating depth as object distance increases, with depth uncertainty growing approximately quadratically with distance. Therefore, LiDAR measurements are preferred for longer-range distance estimation. Significant differences between camera and LiDAR measurements may indicate sensor errors or calibration issues. Misalignment between the two sensors can be detected by projecting LiDAR points onto camera images and corrected through recalibration.

VectorNav 200 : GNSS measurements can become unreliable near buildings, trees and other obstacles due to signal reflections and blockage. This may cause positioning errors or temporary loss of satellite fixes. Although the VN-200 uses inertial measurements to maintain position estimates during short interruptions, accumulated drift reduces accuracy over time. To maintain reliable localization, wheel odometry and LiDAR scan matching can be used to estimate vehicle motion until GNSS measurements become available again.

Wheel Speed Sensor : Wheel slip can cause measured wheel speed to differ from the vehicle's actual speed. This can be identified by comparing wheel speed measurements with IMU, GNSS or LiDAR-based velocity estimates. When slip is detected, the sensor fusion algorithm reduces its reliance on wheel speed data. Incorrect tyre radius or pulses-per-revolution settings can introduce speed estimation errors, which are corrected through measured-distance calibration. Calibration may need to be repeated when tyre specifications, pressure or loading conditions change. At low speeds, measurement accuracy can be improved by calculating the time interval between consecutive pulses. Sensor failure can also be detected when one wheel reports zero speed while the remaining wheels are rotating.

CONCLUSION

Autonomous vehicles depend on the integration of various sensors, computing platforms and electronic control systems for safe and efficient operation. Sensors such as LiDAR, cameras, GNSS, IMU and wheel speed sensors provide essential information about the vehicle's surroundings, position and motion. Sensor fusion improves the accuracy and reliability of this information, while the NVIDIA Jetson AGX Orin processes the data to support perception, navigation and decision-making. Steer-by-wire and brake-by-wire systems convert these decisions into vehicle movements. Understanding the working principles, applications and limitations of these technologies provides a foundation for developing reliable autonomous driving systems.

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